Can Statistical Signal Processing Predict the Bitcoin Price?

A comparison of AR, Wiener, adaptive and Kalman predictors against the random walk

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1 Introduction and motivation

Bitcoin is a decentralised digital currency introduced in 2008 [1]. Bitcoin's daily price changes of several percent make forecasting attractive for traders and for risk management, and the price series is a natural test case for the predictors of this course. The question addressed is: *given the past daily prices, can a linear or adaptive predictor forecast the next day's price better than simply repeating today's price?* That "random-walk" forecast is the standard benchmark in financial forecasting [2], [3], and the efficient-market hypothesis [4] implies that this benchmark is hard to beat. A negative answer also has value, as a warning against predictors that only look good on a short validation period (Sec. 6). Whether the Bitcoin market is efficient is debated in the literature [5]–[8]. In this work, (i) the second-order statistics of the data are analysed to build a data model, (ii) the course predictors are implemented (AR via Levinson–Durbin, FIR Wiener, LMS, NLMS, RLS and two Kalman filters [9], [10]), (iii) the predictors are tuned on a validation block and compared with the random walk on unseen test data with a significance test, and (iv) the test is repeated on recent data (2025–26) and the data model is checked on the full 12-year history. All code is in the accompanying notebook (Python with NumPy, SciPy, statsmodels and Matplotlib [11]–[14]).

2 Data and data model

Data. The course file contains daily Yahoo Finance BTC-USD prices from 6 Aug 2020 to 6 Aug 2021. Every close was compared with the current Yahoo Finance history: 345 of 363 days agree to the cent; 16 days around an October 2020 gap were revised by at most 0.2 % and 28 Jan 2021 by 5.7 %; three days (9, 12, 13 Oct 2020) are empty; and the last row (6 Aug 2021) was saved before the day closed. The course data are kept, the last row is dropped, the gaps are filled by log-linear interpolation, and a filled day is never scored. This leaves $N = 365$ prices (364 returns). The closing price $p[n]$ is predicted via the log-price $x[n] = \ln p[n]$ and the log-return $r[n] = x[n] - x[n-1]$; a return prediction $\hat{r}[n]$ gives $\hat{p}[n] = p[n-1]e^{\hat{r}[n]}$. The returns are split chronologically into three blocks (Fig. 1): *fit* (233), *validation* (28 Mar–24 May 2021, 58) and *test* (25 May–5 Aug 2021, 73). No choice of any kind is made on the test block, and the data-model statistics below use only the 291 pre-test returns.

Statistics (Table 1, Fig. 2). The log-price has a unit root, whereas the returns are stationary in the mean, with a mean not significantly different from zero. The returns are *uncorrelated*: the sample ACF stays within $\pm 1.96/\sqrt{N}$, Ljung–Box [15] does not reject whiteness, and AIC and MDL [16], [17] both select AR order $p = 0$. The returns are *not* i.i.d. Gaussian: the tails are heavy, $r^2[n]$ is autocorrelated, and the variance of the second half of the sample is 2.9 times that of the first. These are the stylised facts of asset returns [18].

Spectrum. For white noise the power spectrum is flat at σ^2. The periodogram (Fig. 2c) is asymptotically unbiased, but the periodogram variance, $\approx P^2(e^{j\omega})$, does not decrease with N; each ordinate scatters over roughly -16 to $+6$ dB. Bartlett's method ($K = 8$ rectangular segments of $L = 36$, resolution $0.89/L = 0.025$ cycles/day) and Welch's method [21] (Hann window, $L = 64$, 50 % overlap, $K = 8$; course variance $\approx \frac{9}{8K}P^2$ for a Bartlett window, slightly more for Hann, resolution $1.44/L = 0.022$ cycles/day) trade resolution for variance. That easily resolves a weekly line at $1/7$ cycles/day. Both estimates stay near 0 dB and show no peak; 6 % of the Welch bins fall outside the approximate 95 % band (5 % expected). The AR(4) spectrum deviates by less than ± 1.5 dB.

Data model. The log-price is modelled as a random walk driven by the returns,

$$x[n] = x[n-1] + r[n], \qquad r[n] = \sum_{k=1}^{p} a_k\, r[n-k] + e[n], \tag{1}$$

where the innovation $e[n]$ is zero-mean and uncorrelated with the past. The random parameters are the $e[n]$; the coefficients $\boldsymbol{a}$ are unknown and may drift. The returns $r[n]$ are treated as (locally) wide-sense stationary in the second-order structure, with a conditional variance that changes in time (GARCH-type); the F-test

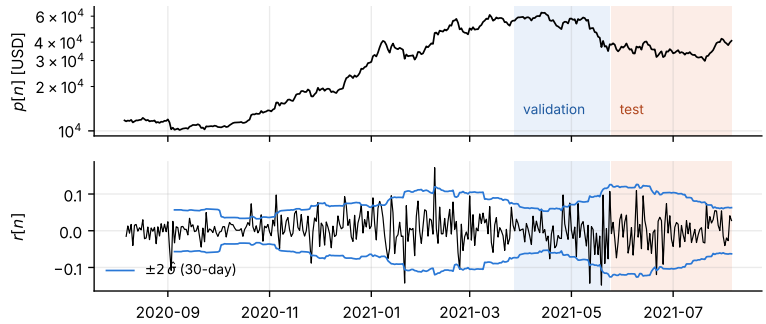


Figure 1: Closing price (log scale) and daily log-return $r[n]$ with ± 2 rolling standard deviations (30 days). Validation block in blue, test block in orange. The validation block contains the April 2021 peak and the May 2021 crash.

Table 1: Statistics of the 291 pre-test log-returns (7 Aug 2020 – 24 May 2021).

Property	Test / estimate	Result
Mean, std	$\bar{r}$, $\hat{\sigma}$; t-test $\bar{r} = 0$	0.0041, 0.0400; $p = 0.08$
Shape	skewness, excess kurtosis; Jarque–Bera	-0.16, 2.55; $p \approx 4 \cdot 10^{-18}$ (not Gaussian)
Unit root in $x[n]$	ADF [19] (const./trend); KPSS [20]	$p = 0.75$ / 0.99 (unit root); $p \leq 0.01$
Stationarity of $r[n]$	ADF; KPSS	$p < 0.001$; $p \geq 0.10$ (stationary in the mean)
Whiteness of $r[n]$	Ljung–Box, lag 5 / 10 / 20	$p = 0.42$ / 0.63 / 0.64 (white)
Volatility clustering	Ljung–Box on $r^2[n]$, lag 5 / 10 / 20	$p = 0.49$ / 0.030 / 0.006
Constant variance	F-test, first vs second half	std 2.8 % vs 4.9 %, $p \approx 10^{-10}$
AR order	AIC, MDL ($p \leq 20$)	both select $p = 0$

above assumes i.i.d. data and only indicates this variation, so strict WSS over the whole sample is an approximation. Estimating the ACF and spectrum from one record assumes ergodicity. The minimum-MSE predictor is the conditional mean $E\{r[n] \mid r[n-1], \dots\}$. If the returns are white ($\boldsymbol{a} \approx \mathbf{0}$), the best *linear* predictor is the (insignificant) mean, i.e. $\hat{r}[n] \approx 0$ and $\hat{p}[n] = p[n-1]$. The models below test whether a small or slowly varying $\boldsymbol{a}$ can still be exploited. The whiteness tests tend to over-reject under volatility clustering, so "not rejected" is a conservative finding.

3 Methodology

All predictors are causal ($\hat{r}[n]$ uses only $r[1], \dots, r[n-1]$), were implemented from scratch, and were checked against each other and against statsmodels.

Baselines. The *naive* random walk $\hat{r}[n] = 0$, and a *drift* model $\hat{r}[n] =$ mean of past returns.

AR(p) / Wiener. The Yule–Walker equations $\mathbf{R}_r \boldsymbol{a} = \boldsymbol{r}_r$ use the biased autocorrelation estimate (positive definite, so the model is stable). The equations are solved by Levinson–Durbin [9], [22] and re-estimated every day on all past data. The FIR Wiener one-step predictor ($d[n] = r[n]$, inputs $r[n-1], \dots, r[n-L]$) satisfies $\mathbf{R}_r \boldsymbol{w} = [r_r(1), \dots, r_r(L)]^T$, which *are* the Yule–Walker equations; solving both independently gave $\max |\boldsymbol{w} - \boldsymbol{a}| < 10^{-16}$. On the fit block the prediction gain $G = r_r(0)/\xi_{\min}$ is at most 0.08 dB ($L = 8$): less than 2 % lower error power, even in-sample. ARMA(1,1) was also fitted once by maximum likelihood [23].

Adaptive filters [9], [24]. With $\boldsymbol{\phi}[n] = [r[n-1], \dots, r[n-p]]^T$ and a-priori error $e[n] = r[n] - \boldsymbol{w}^T[n]\boldsymbol{\phi}[n]$:

$$\text{LMS:} \quad \boldsymbol{w}[n+1] = \boldsymbol{w}[n] + \mu\, e[n]\, \boldsymbol{\phi}[n], \qquad \mu = c \cdot 2/(p\hat{\sigma}^2), \tag{2}$$

$$\text{NLMS:} \quad \boldsymbol{w}[n+1] = \boldsymbol{w}[n] + \tilde{\mu}\, e[n]\, \boldsymbol{\phi}[n]/(\epsilon + \|\boldsymbol{\phi}[n]\|^2), \qquad 0 < \tilde{\mu} < 2,\ \epsilon = p\hat{\sigma}^2 \approx E\|\boldsymbol{\phi}\|^2, \tag{3}$$

$$\text{RLS:} \quad \boldsymbol{k}[n] = \frac{\mathbf{P}[n-1]\boldsymbol{\phi}[n]}{\lambda + \boldsymbol{\phi}^T[n]\mathbf{P}[n-1]\boldsymbol{\phi}[n]}, \quad \boldsymbol{w}[n] = \boldsymbol{w}[n-1] + \boldsymbol{k}[n]e[n], \tag{4}$$

$$\mathbf{P}[n] = \lambda^{-1}\big(\mathbf{P}[n-1] - \boldsymbol{k}[n]\boldsymbol{\phi}^T[n]\mathbf{P}[n-1]\big), \qquad \mathbf{P}[0] = \hat{\sigma}^{-2}\mathbf{I}. \tag{5}$$

The course gives $0 < \mu < 2/\mathrm{tr}(\mathbf{R}_r) = 2/(p\sigma^2)$ as the condition for mean-square convergence of LMS (under the independence assumption), so the LMS step is a fraction c of that bound. RLS has an effective memory of $1/(1-\lambda)$ samples.

Kalman filters. (i) *Time-varying AR* (TV-AR). State $\boldsymbol{a}[n] = \boldsymbol{a}[n-1] + \boldsymbol{w}[n]$, $\boldsymbol{w} \sim \mathcal{N}(\mathbf{0}, q\mathbf{I})$; observation $r[n] = \boldsymbol{\phi}^T[n]\boldsymbol{a}[n] + v[n]$, $v \sim \mathcal{N}(0, \sigma_v^2)$. For $q = 0$ (with matching initialisation) the filter reduces to RLS with $\lambda = 1$. (ii) *Local level*, a state-space model of the price (Xu and Zhang [25] also model a price in state-space

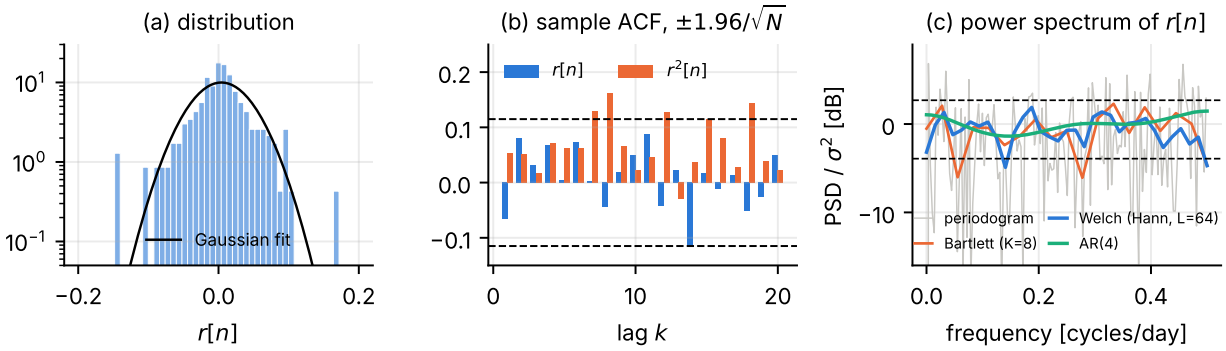


Figure 2: Pre-test returns. (a) Histogram (log density) with a Gaussian fit: heavy tails. (b) Sample ACF of $r[n]$ and $r^2[n]$ with the $\pm 1.96/\sqrt{N}$ bound. (c) Periodogram, Bartlett, Welch and AR(4) spectra normalised by σ^2; dashed: approximate 95 % interval of the Welch estimate for a flat spectrum (χ^2 with $2 \cdot 8K/9$ degrees of freedom).

Table 2: One-step test results, 25 May – 5 Aug 2021 (73 days). Rel. MSE: return MSE divided by that of the naive predictor. DM: statistic (positive = worse than naive) and p; Holm: multiplicity-adjusted p.

Model	RMSE [$]	MAE [$]	MAPE [%]	Dir. acc. [%]	Rel. MSE	DM (p)	Holm p
Naive (random walk)	**1454**	**1143**	**3.21**	–	1.000	–	–
Drift (mean)	1462	1144	3.22	53.4	1.008	0.54 (0.59)	1.00
AR(4)	1484	1172	3.29	47.9	1.039	1.30 (0.20)	0.80
ARMA(1,1)	1460	1152	3.23	49.3	1.012	0.52 (0.61)	1.00
LMS ($p = 4$)	1588	1244	3.48	45.2	1.179	2.32 (0.02)	0.14
NLMS ($p = 2$)	1539	1210	3.39	42.5	1.117	2.56 (0.01)	0.10
RLS ($p = 4$, $\lambda = 0.97$)	1606	1263	3.53	47.9	1.205	2.18 (0.03)	0.16
Kalman TV-AR ($p = 4$)	1619	1274	3.56	43.8	1.225	2.40 (0.02)	0.13
Kalman local level	1478	1165	3.26	45.2	1.025	0.90 (0.37)	1.00

form). $s[n] = s[n-1] + w[n]$, $x[n] = s[n] + v[n]$, prediction $\hat{x}[n] = \hat{s}[n-1|n-1]$. In steady state this is exponential smoothing with gain K_∞; $K_\infty \to 1$ recovers the random walk. Without the Gaussian assumption the Kalman filter is still the best *linear* estimator. With $\sigma_v^2 = \hat{\sigma}^2$ fixed, q (or q/σ_v^2) is tuned.

Protocol. All hyper-parameters minimise the one-step MSE of r on the validation block, over 130 grid points: AR with $p = 0..10$ ($p = 0$ is the naive predictor); LMS, NLMS, RLS and TV-AR with $p \in \{1, 2, 4, 8\}$ and, respectively, $c \in [0.001, 0.3]$, $\tilde{\mu} \in [0.001, 0.3]$, $\lambda \in [0.9, 1]$, $q \in [10^{-6}, 10^{-2}]$; and the local level with $q/\sigma_v^2 \in [10^{-3}, 10^{4}]$. The test block is then predicted one step at a time. Reported are the RMSE, MAE and MAPE of the price (the error level reflects volatility; only the ratio to naive measures skill), the return MSE relative to naive, directional accuracy (tested against the 53.4 % share of up-days, binomial), and the Diebold–Mariano (DM) test [26] on squared *price* errors, with $d[n] = e^2_{\text{model}}[n] - e^2_{\text{naive}}[n]$ and statistic $\sqrt{(T-1)/T}\,\bar{d}/\sqrt{\hat{\gamma}_d(0)/T}$ (small-sample factor of [27]) against t_{T-1}. Because several models are tested, Holm-adjusted p-values are also given. The random walk is nested in the AR, RLS and Kalman models, a case for which dedicated tests exist [28], so the DM p-values are read as approximate.

4 Results

Validation. Validation selects AR(4); LMS ($p = 4$, $c = 0.03$); NLMS ($p = 2$, $\tilde{\mu} = 0.1$); RLS ($p = 4$, $\lambda = 0.97$); TV-AR ($p = 4$, $q = 10^{-3}$); and the local level ($q/\sigma_v^2 = 3.2$, $K_\infty \approx 0.80$). On the validation block the adaptive filters reduce the MSE by about 11 % relative to naive, and AR(4) by 2 % (Fig. 4a). Even there, none of these gains is significant (DM $p = 0.09$–0.40).

Test. Table 2 gives the main result: *no model beats the random walk*; every point estimate is worse.

- *Tied:* drift, ARMA(1,1), AR(4) and the local-level filter are within 1–4 % (DM $p \geq 0.20$). For AR(4) the price-error MSE changes by +4.2 %, 95 % CI $[-2.1, +10.6]$ %.
- *Worse:* the four adaptive filters that won on validation are 12–23 % worse. This is individually significant ($p = 0.01$–0.03), but not after the Holm correction ($p = 0.10$–0.16).
- *Direction:* no model predicts the sign better than the up-day base rate (binomial $p \geq 0.07$).
- *Residuals:* no model leaves significant residual autocorrelation (Ljung–Box $p \geq 0.38$). The fitted

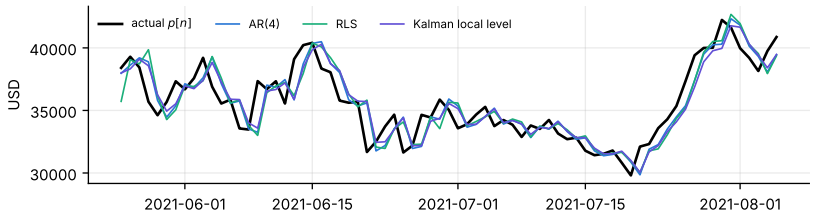


Figure 3: Actual price and one-step predictions on the test block: all follow the previous day's price.

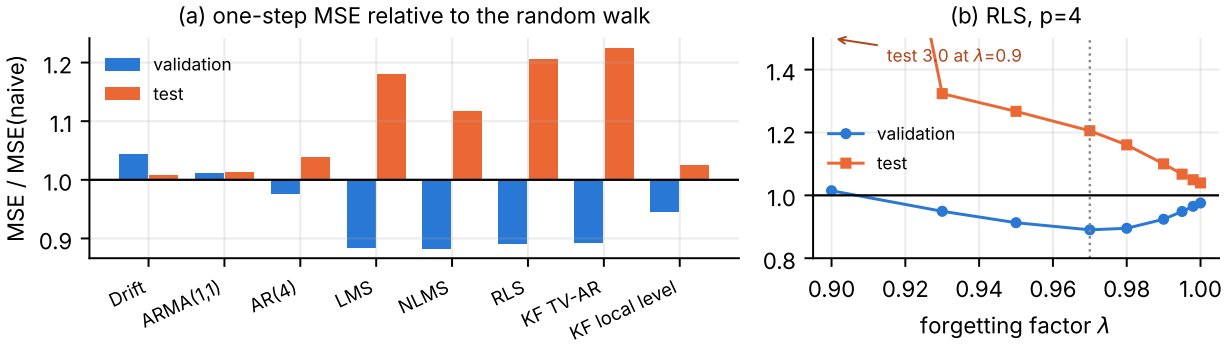


Figure 4: (a) One-step MSE relative to the random walk on the validation block (used for tuning) and the test block; below 1 beats the random walk. (b) RLS ($p = 4$): relative MSE versus λ; dotted line: value chosen on validation.

ARMA(1,1) pole (0.994) and zero (0.978) nearly cancel: maximum likelihood chose an almost-white model.

For the tied models the test can only detect MSE differences of about 5–10 % at 80 % power (13–28 % for the adaptive filters), while whiteness limits any linear gain to below 2 %. The conclusion therefore rests on the data model plus consistent point estimates, not on the DM test alone (Fig. 3: all forecasts essentially follow the previous day's price).

Why validation and test disagree. Selection alone explains little: on 40 shuffled copies of the returns (all structure removed), the best of the 28 RLS settings is at best 0.94 (mean 1.00), against 0.89 on the real validation block. Instead, there is evidence of temporary structure in the validation block (the April peak and May 2021 crash, Fig. 1). This block shows lag-1 and lag-2 autocorrelations of -0.31 and $+0.27$, outside the ± 0.26 bound (Ljung–Box $p = 0.035$, which may partly reflect the crash's volatility), whereas the fit and test blocks are white (Ljung–Box $p = 0.88$ and 0.54). The adaptive filters tracked this structure, which vanished in the test period. Fig. 4b shows the consequence for RLS: every $\lambda < 1$ that helps on validation hurts on test, and the test error approaches the naive error only as $\lambda \to 1$, i.e. almost no adaptation.

Longer horizons, other prices and recent data (Table 3). *Horizons:* errors of 5- and 10-day forecasts grow roughly like $\sqrt{h}$, as for a random walk; only the local-level filter is marginally below the random walk (by < 0.5 %, untested). *High and low prices:* again no model beats naive. *Open price:* because Bitcoin trades 24/7, Open$[n]$ equals Close$[n-1]$ up to an RMSE of \$56, so the open price is not a meaningful target. *Recent data* (test 1 Aug – 15 Oct 2026, 76 days): the returns are again white (Ljung–Box $p = 0.30$, lag 10) with strong volatility clustering ($p < 10^{-4}$ on r^2). Re-tuned on these data, AR selects $p = 0$ and the local-level filter $K_\infty \approx 1$: both reduce to the random walk. NLMS is the closest competitor (rel. MSE 0.998; 0.97 and 0.96 for high and low), but far from significant (DM $p \geq 0.69$). *Frozen 2021 settings*, the stricter out-of-sample test. All models are 2–18 % worse; AR(4) ($p = 0.03$) and LMS ($p = 0.05$) are individually significant, but not after the Holm correction ($p \geq 0.23$).

Full-history robustness. The full history (17 Sep 2014 – 15 Oct 2026) was not used for the main analysis, because the brief provides one year for modelling and newer data for evaluation, the market changed (volatility per year ranges from 2.2 to 4.9 %), and the literature reports that Bitcoin's efficiency changes over time [5], [7]. Pooling would therefore mix different regimes. Instead, the full history is used to check the data model year by year (Table 4). Whiteness is rejected in only 3 of 12 years: 2015 and 2016, and 2020 with the March COVID crash. The heteroskedasticity-robust variance ratio of Lo and MacKinlay [29] is never significant ($|z^*| \leq 1.28$ for $q = 5$ and 20). That test is the "price-chart" view of the same statistics: $VR(q) > 1$ means multi-day trends, $VR(q) < 1$ mean reversion. Volatility clustering, in contrast, is significant in 7 of 12 years. Over all 4411 days a lag-1 correlation of -0.022 is detectable (Ljung–Box, lag 10, $p = 0.03$), but far too small to exploit.

Table 3: Left: RMSE [$] of h-step forecasts (2021). Middle: one-step RMSE [$], high and low prices (2021). Right: recent test block (76 days), settings re-tuned on recent data or frozen from 2021.

	2021, horizon h			2021, $h = 1$		2026 re-tuned		2026 frozen	
Model	1	5	10	High	Low	Rel. MSE	DM p	Rel. MSE	DM p
Naive (random walk)	1454	3088	4080	1315	1382	1.000	–	1.000	–
AR(p^*) ($p^* = 4$; 0 re-tuned)	1484	3160	4185	1322	1399	1.000	–	1.036	0.03
NLMS	–	–	–	1365	1485	0.998	0.95	1.018	0.48
RLS	1606	3297	4232	1397	1536	1.084	0.52	1.183	0.27
Kalman TV-AR	1619	3296	4207	1398	1535	1.081	0.56	1.142	0.27
Kalman local level	1478	3080	4062	1341	1468	1.000	0.75	1.098	0.30

Table 4: Daily log-returns per year (2026 to 15 Oct). $\hat{\sigma}$ in %/day; LB: Ljung–Box p, lag 10; robust z^* in brackets; "all" includes Sep–Dec 2014.

	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	all
$\hat{\sigma}$	3.7	2.5	4.9	4.3	3.5	4.0	4.2	3.4	2.3	2.8	2.2	2.4	3.5
$\hat{\rho}(1)$	.00	−.02	.03	−.04	−.03	−.13	−.05	−.01	.01	−.06	−.09	−.03	−.02
LB r	<.001	.005	.73	.31	.29	.005	.38	.32	.72	.57	.82	.19	.03
LB r^2	<.001	<.001	<.001	<.001	<.001	.68	.12	.23	.14	.12	<.001	<.001	<.001
VR(20)	.71	.87	.96	.98	1.06	.95	.97	.85	1.13	1.00	.73	1.23	1.09
(z^*)	(−.8)	(−.3)	(−.1)	(−.1)	(.2)	(−.1)	(−.1)	(−.5)	(.5)	(.0)	(−.9)	(.6)	(.9)

5 Discussion

Interpretation. The results agree with the data model. The returns are white, so the Wiener–Hopf solution is $\boldsymbol{w} \approx \mathbf{0}$, and fixed-coefficient predictors gain less than 2 % even in-sample. Out-of-sample, the estimation noise in $\boldsymbol{a}$ costs more than that, so each is slightly worse than $\boldsymbol{w} = \mathbf{0}$. This is consistent with weak-form market efficiency [4]. Time-varying predictors are not bound by this gain and did exploit the validation block's temporary structure. Because such structure does not persist, short memory mainly adds variance (the tracking/misadjustment trade-off); the excess MSE of 12–23 %, far above the misadjustment expected under stationarity (about 3 % for LMS, $M \approx c$; about 6 % for RLS, $M \approx p(1-\lambda)/(1+\lambda)$), reflects weights carrying validation-period structure into the test block, changing input power and heavy-tailed outliers. The local-level filter is robust only when re-tuned (the filter then collapses to the random walk); with the 2021 gain frozen, the filter is 10 % worse on recent data.

Comparison with the literature. The AR/ARMA results agree with [30], where ARIMA performed poorly and the best LSTM reached only 52 % directional accuracy, and with [31], where ARIMA was useful only over short intervals and depended on the period. The Kalman study in [25] reports effective predictions on 27 closing prices. On a sample that short, the validation block of this study would also have suggested a gain; the same holds for choosing the ARMA order by next-day error [32]. As in [5], [7], return dependence is found to be concentrated in specific periods and, as in [7], volatility dependence is much more persistent. Gradojevic et al. [3] beat the random walk at the daily horizon, but with nonlinear models and technical indicators, and Jang and Lee [33] add blockchain data. Both are outside the linear, price-only framework.

Pros and cons. AR/Wiener: cheap, interpretable and stable, but static. LMS: simple, $O(p)$, but sensitive to the step size and to input power (NLMS fixes the latter). RLS: fast, memory set by λ, but $O(p^2)$ and tracks noise. Kalman: explicit noise model, contains RLS ($\lambda = 1$) and the random walk, but needs q, σ_v^2.
Limitations. Short test blocks (limited power); constant noise variance assumed; only past prices used.

6 Conclusions and future work

Daily Bitcoin log-returns are, to second order, close to white noise (2020–21, 2025–26, and 9 of 12 years since 2015). Accordingly, no course predictor beats the random walk; the best are within about 1 %. The adaptive filters looked 11 % better on validation but are 12–23 % worse on test, most likely by tracking temporary structure around the May 2021 crash. On recent data, re-tuning drives AR and the local-level filter to the random walk. The persistent structure is in the *variance*: next steps are GARCH-type models [34], [35] (forecasting risk, not price), a Kalman filter with time-varying noise variance, and nonlinear models with exogenous inputs [3], [33], always tested against the random walk.

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